

Ali Ataollahi

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EDUCATION

B.Sc. Computer Engineering

2020 - 2025

Department of ECE, University of Tehran, Tehran, Iran

Cumulative GPA: 17.88/20 (3.84/4.00)

High School Diploma

2017 - 2020

Shahid Soltani 4 High School, Karaj, Iran

GPA: 19.66/20 (4.00/4.00)

RESEARCH INTERESTS

- Symbolic and Inductive Model Checking
- Program Analysis and Semantics
- Verification under Program Evolution
- Process Algebra and Actor Models
- Cyber-Physical Systems and Digital Twins
- Interactive and Automated Theorem Proving

PUBLICATIONS

Extending MoXChecker

Nian-Ze Lee, Salih Ates, Po-Chun Chien, **Ali Ataollahi**, Marziyeh Mousavi, Dirk Beyer
in preparation

Proving Timed Weak Equivalence of Actors based on Slice Equivalence

Ali Ataollahi, Fatemeh Ghassemi, Eduard Kamburjan, Marjan Sirjani
in preparation

Reachability analysis of Hybrid Rebeca models

Saeed Zhiany, Fatemeh Ghassemi, Nesa Abbasimoghadam, Ali Hodaei, **Ali Ataollahi**, József Kovács, Erika Ábrahám, Marjan Sirjani
published in Journal of Systems Architecture, available at [this link](#)

Generalizing Featured Finite State Machines Constructed by Model Merging

Shaghayegh Tavassoli, Ramtin Khosravi, **Ali Ataollahi**
submitted to SoSyM, preprint available at [this link](#)

Smart Contracts Verification using mCRL2

Nazanin Torabi Goudarzi, Fatemeh Ghassemi, **Ali Ataollahi**, Hossein Hojjat
in preparation

RESEARCH EXPERIENCE

Software Systems Lab (SoSy-Lab), LMU Munich

MoXChecker — *Symbolic Model Checking*

- Implemented and optimized IC3/PDR components and improved IC3IA correctness and performance, including invariant handling, bit-vector counterexamples, generalization, propagation, and bounded-model-checking lookahead.
- Integrated direct MoXI input and subsystem flattening, refactored verification engines into a modular architecture, and corrected QF_BV parsing and model-loading failures.
- Built BenchExec configurations and reproducible studies comparing BMC, k -induction, IMC, ISMC, IC3, and IC3IA across theories, encodings, SMT solvers, and interpolation backends.
- Investigated how background theories and solver features change algorithm rankings, how complementary the engines are, and how MoXI tools compare with CHC-, VMT-, and Btor2-based verification flows.

Adaptive Programming Research Collaboration

AdaptChecker — Verification Reuse for Evolving Timed Actor Models

- Co-designed a static criterion that compares observable slices of Rebeca Dependence Graphs and guarantees weak timed bisimulation, enabling reuse of an existing tiny twin after behavior-preserving model changes.
- Implemented AdaptChecker end to end in Python: Timed Rebeca lexer/parser, six-kind dependence-graph construction, backward slicing, typed graph-isomorphism checking, diagnostics, and DOT visualization.
- Resolved discrepancies between the formal definitions and worked examples through explicit compatibility modes and regression tests; created 48 self-tests covering examples and case studies.
- Evaluated ten model/revision pairs: every static decision completed in under 44 ms and 23 MiB, while regeneration for the hardest case took over an hour and hundreds of GiB.

University of Tehran, Formal Methods and System Verification Lab

- **Hybrid Rebeca:** Contributed to direct bounded reachability analysis for asynchronous CPS models with discrete software behavior and continuous dynamics.
- **Smart-Contract Verification:** Built abstractions and translation steps from Solidity to mCRL2 models for detecting vulnerability-specific behaviors through model checking.

Soren — Remote Static-Analysis Collaboration

- Developed a Joern plugin for discovering Java-microservice API attack surfaces, handling annotations and reflection in code-property graphs, producing OpenAPI artifacts, and exposing the analysis through Docker services.

University of Tehran, Dependable Systems Design Lab

- **Real-Time Object Detection:** Built a dynamic-deadline YOLOv8 pipeline using pruning, quantization, layer skipping, and dynamic path selection for resource-constrained CPS.

COURSERA ONLINE COURSES

System Validation: Automata and Behavioural Equivalences

Eindhoven University of Technology — Instructor: [Jan Friso Groote](#)

Automated Reasoning: Symbolic Model Checking

Eindhoven University of Technology — Instructor: [Hans Zantema](#)

Automated Reasoning: Satisfiability

Eindhoven University of Technology — Instructor: [Hans Zantema](#)

Blockchain Specialization: Smart Contracts

University at Buffalo — Instructor: [Bina Ramamurthy](#)

Programming Languages

Korea Advanced Institute of Science and Technology(KAIST) — Instructor: [Sukyoung Ryu](#)

TEACHING EXPERIENCE

Compiler Design Tutorial Series

2024–Present

[YouTube Channel](#)

TEACHING ASSISTANT EXPERIENCE

- **Compiler and Programming Languages Design** — *Spring and Fall 2024*. Head Teaching Assistant for [Prof. Fatemeh Ghassemi](#) (Fall) and Dr. Shaghayegh Tavassoli (Spring); led the TA team, designed assignments and educational languages, developed compiler projects, and delivered tutorials and additional sessions.
- **Operating Systems** — *Fall 2023–2024*. Assignment designer and lecturer assistant for [Dr. M. Kargahi](#); prepared systems-programming exercises and supported lectures.

- **Database Design** — *Spring and Fall 2024*, with [Dr. A. Shakery](#).
- **Data Structures** — *Spring 2022–Fall 2024*, with [Dr. H. Faili](#).
- **Advanced Programming** — *Spring 2022–2024*, with [Prof. R. Khosravi](#).
- **Engineering Statistics** — *Fall 2022–2023*, with [Dr. B. Bahrak](#) and [Dr. M. Tavassolipour](#).
- **Computer Networks** — *Fall 2023*, with [Dr. A. Khonsari](#).
- **Computer-Aided Design** — *Fall 2023*, with [Dr. M. Ersali](#) and [Prof. M. Modarressi](#).
- **Operations Research** — *Fall 2023*, with [Dr. A. Ramezani Moghadam](#).

SELECTED TECHNICAL PROJECTS

- **Formal Verification with Isabelle/HOL:** Formalized safety properties of hybrid and dynamic systems and exercised automated and interactive proof strategies.
- **FunctionCraft Language Compiler:** Built an ANTLR4-based educational compiler covering parsing, AST construction, functional-language features, and JVM bytecode generation.
- **IoT Access-Control System:** Led the C++/Qt server implementation, including HTTP and WebSocket communication between RFID hardware and the graphical interface.

HONORS AND AWARDS

- Among top 15% of B.Sc. students in Computer Engineering at University of Tehran.
- Ranked 89th (top 0.1%) among almost 165,000 participants in the Nationwide Iranian University Entrance Exam in Mathematics and Physics.

SKILLS

- **Formal Methods:** symbolic model checking, IC3/PDR, IC3IA, BMC, k -induction, interpolation, SMT, reachability analysis, process algebra, behavioral equivalence, and theorem proving.
- **Programming Languages:** experienced in C, C++, Python, Java, and Scala; familiar with Go, Bash, Solidity, and mCRL2.
- **Verification and Analysis Tools:** Z3, PySMT, Isabelle/HOL, Coq, BenchExec, Joern, ANTLR, Lark, Graphviz, and $\text{\LaTeX}$.
- **Engineering:** Git, Docker, Linux, experimental evaluation, reproducible benchmarking, and technical writing.